AI ENGINEERING / LOGISTICS / FINANCE OPERATIONS

# FREIGHT INVOICE AUDIT

08

PROPOSED RESEARCH / NOT YET VERIFIED

THE ENGINEERING QUESTION

**How much of a carrier overcharge can deterministic recalculation find and prove without raising disputes that finance would reject?**

## PROBLEM DEFINITION AND RESEARCH CONTRIBUTION

A proposed audit engine that recalculates carrier invoices from contract rates, fuel surcharges, accessorials and delivery events then reports exact overcharges with evidence.

- Recalculate from contract terms so every reported pound has a traceable rule behind it.
- Treat ambiguous clauses as questions for a person rather than guesses made by a model.
- Use synthetic carrier data only and never imply recovered money or live customer results.

## ARCHITECTURE AND IMPLEMENTATION SPECIFICATION

1 STAGE **Invoice intake** → 2 STAGE **Contract rating** → 3 STAGE **Delivery events** → 4 STAGE **Finance review**

- Typed contract and rate card store with effective dates, zones, weight breaks, minimums and fuel surcharge tables.
- Rating engine that rebuilds each charge line from shipment and delivery events then diffs it against the invoice line.
- Evidence pack per variance: rule applied, inputs used, expected amount, billed amount and a person's decision record.

### ENGINEERING DECISIONS TO MAKE EXPLICIT

The design review must record data contracts, state transitions, failure handling and why the selected baseline is defensible. Every subsystem should have an interface specification and a reproducible test route.

## DESIGN BOUNDARIES AND ASSUMPTIONS

⚠ Synthetic contracts cannot show how real tariffs or real carriers behave in disputes.

⚠ A recalculated difference is a recommendation and may reflect a missing amendment.

ⓘ Scope is proposed unless explicitly identified as documented existing work. All measured claims require reproducible evidence.

FREIGHT INVOICE AUDIT / PROTOCOL / ACCEPTANCE / REVIEW

# EXPERIMENTS, VERIFICATION & PUBLICATION

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## EXPERIMENTAL METHODOLOGY

01 Generate synthetic contracts, shipments, delivery events and invoices with labelled overcharges and wrong surcharges.

02 Compare manual style sampling, rules only rating and a hybrid using a model solely to read unstructured clauses.

03 Measure variance recall, exact amount accuracy, false disputes and reviewer time per invoice on held out synthetic batches.

## MEASUREMENT AND ACCEPTANCE EVIDENCE

- Overcharge found vs injected truth on synthetic invoices
- False dispute rate and unexplained variance by cause
- Examples where rate cards are ambiguous or contradictory

## REQUIRED REPOSITORY ARTEFACTS

- Synthetic carrier invoice and contract dataset
- Rating engine with variance test suite
- Evidence pack template and evaluation report

### TECHNICAL REVIEW QUESTIONS

**Which assumptions could invalidate the result? How do the baselines behave under the same conditions? What is the strongest negative case? Which variables, seeds, workloads, software versions and machine properties must be retained for independent reproduction?**

### VERIFICATION MAPPING

For each acceptance claim, record the test identifier, input configuration, observed result, pass/fail definition and an independent rerun procedure. Failed or inconclusive cases remain in the report rather than being silently excluded.

### PUBLICATION DISCIPLINE

Release original source, configuration files, test harnesses, raw measurements and scripts where rights permit. Distinguish simulation outcomes from physical validation and completed achievements from planned evaluation. Results must not be inferred from targets.